Customer Segmentation & Repeat-Purchase Prediction

Project Report — Vaishnavi Lahoti

1. Objective

Group customers by buying behavior and predict repeat buyers to support targeted marketing.

2. Project Overview

The project uses RFM-style features to understand customer purchasing behavior, K-Means clustering with four segments, and Logistic Regression for repeat-purchase prediction. The documented project description reports elbow and silhouette validation for k=4 and approximately 80% classification accuracy.

3. Analytical Workflow

1. Prepare and inspect customer-level data
2. Create RFM-style features
3. Validate K-Means clustering (k=4)
4. Build Logistic Regression repeat-buyer model
5. Visualize segments and repeat-purchase behavior
6. Translate findings into targeted-offer recommendations

4. Tools & Techniques

Python, Pandas, NumPy, Scikit-learn, Matplotlib and Seaborn.

5. Key Project Points

• Four customer segments are used to distinguish purchasing behavior.

• Repeat-purchase prediction is treated as a classification problem.

• Visual outputs support targeted marketing decisions.

• The reported ~80% accuracy should be revalidated on the original dataset if recovered.

6. Repository Files

File	Purpose
Customer_Segmentation_Data.xlsx	Analysis-ready customer dataset
Dashboard_Screenshot.png	Main visual preview
Repeat_Purchase_Chart.png	Repeat-purchase visual
Customer_Segmentation_Project_Documentation.pdf	Detailed documentation
README.md	GitHub overview

7. Data Availability Note

The original raw dataset was reported as 3,900 customers but was unavailable during reconstruction. The Excel file in this repository is a reconstructed 3,900-record portfolio dataset and is not represented as the original source file.

Prepared for portfolio / GitHub presentation.